

Investigating Domain Shift in Deep CNN Models for Pneumonia Detection Across Kaggle and RSNA Chest X-Ray Datasets

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Abstract—Latent learning networks have attained a high precision in automated pneumonia identification based on chest X-ray images. Nevertheless, majority of the current research measures performance using a single dataset thereby restricting insight on how models perform when the domain shifts. This study methodically examines cross-dataset robustness of three popular convolutional neural network architectures, including VGG16, ResNet50, and DenseNet121, on two heterogeneous datasets of chest X-rays: Kaggle and RSNA. The four experimental settings were assessed such as in-domain (Kaggle→Kaggle, RSNA→RSNA) and cross-domain (Kaggle→RSNA, RSNA→Kaggle) settings. All the models were now fine-tuned to the same training settings as it would be fairly compared. Findings indicate that performance is severely deteriorated in case of domain shift, especially when Kaggle to RSNA. DenseNet121 has the greatest cross-domain robustness and its accuracy has relatively consistent stability across datasets. Additionally, Grad-CAM images demonstrate how attention patterns in the models change across cross-domain assessment, which demonstrates the difference in features between datasets. The findings underscore the relevance of cross-dataset validation of medical imaging AI and highlight the potential dangers of using in-domain metrics of performance before clinical implementation.

Index Terms—Pneumonia Detection, Chest X-ray, Deep Learning, CNN, Domain Shift, Cross-Dataset Generalization, Transfer Learning, Grad-CAM, Medical Image Analysis, Model Robustness

I. INTRODUCTION

Pneumonia has been a major cause of morbidity and mortality in the global population especially in children and the elderly. One of the most popular diagnostic tools that are used to detect the pneumonia is the Chest X-ray imaging, as it is cost-effective and highly accessible. Nonetheless, the manual interpretation of the chest radiographs should have experienced radiologists and could be prone to inter-observer variation. Deep learning has been incredibly successful in performing medical image classification tasks, such as pneumonia detection in recent years and mostly using Convolutional Neural Networks (CNNs).

It has been shown that transfer learning using pre-trained CNN architectures including VGG16, ResNet50, and DenseNet121 have high classification accuracy on large-scale chest X-ray data when trained and tested. Most studies

demonstrate that performance measures are very high in cases where models are trained and evaluated on the same dataset. Although the results are encouraging, they are not always applicable in clinical practice in a real-world environment. Practically, medical imaging data differs considerably across hospitals because some hospitals have different acquisition devices, pipelines, demographic distributions, and annotation processes. These differences add what is generally known as domain shift—a difference between the distribution of the training data and the deployment data.

Medical AI systems face a major challenge of domain shift. A model that was trained on a set of data can lose its performance by a significant margin when tested on a different set of data with different properties. Although its significance, cross-dataset evaluation has not been studied extensively in the literature on pneumonia detection, most of the studies focusing on this topic tend to maximize in-domain performance. Therefore, not much is known about the extent to which popular CNN architectures can generalize across mixed datasets of the chest X-rays.

Besides performance evaluation, interpretability is a very important requirement in medical imaging applications. Automated diagnostic systems need to be transparent to gain the trust of the clinicians. Methods like Gradientweighted Class Activation Mapping (Grad-CAM) allow visualizing the areas of attention of the model, which reveals how the model views different areas of the image to make the decision towards classification. Nevertheless, minimal efforts have been expended in examining the domain shift conditions of model attention patterns.

It is based on these gaps that this research explores the cross-dataset generalization of three commonly used CNN structures, namely VGG16, ResNet50, and DenseNet121, applied to two heterogeneous datasets in the domain of the chest X-ray aka the Kaggle Chest X-Ray and the RSNA Pneumonia Detection datasets. There are four experimental conditions that are carefully considered: Kaggle→Kaggle, Kaggle→RSNA, RSNA→RSNA, and RSNA→Kaggle. This work will provide an evaluation of architectural robustness and interpretability with domain shift by quantifying the degradation of perfor-

mance and visualizing Grad-CAM of architecture on the basis of domain changes.

The main contributions of this work are as follows:

- A comparative cross-dataset analysis of three pretrained CNN models to detect pneumonia.
- The domain shift impact was occurring on four experimental settings and measured quantitatively.
- Comparative study of the architectural robustness, where DenseNet121 is found the most robust across the domain transfer.
- Examining attention behavior across domains through Grad-CAM-based interpretability analysis.

The results presented in this study indicate the importance of cross-dataset validation in medical imaging AI and prove that high in-domain accuracy is not equivalent to reliable generalization. This information can be used to further the application of deep learning models in clinical practice in a safe and robust way.

II. PROBLEM STATEMENT

Despite the high accuracy reported in the classification of the deep learning-based pneumonia detection system on the benchmark chest X-ray data sets, the majority of the current research papers analyze the accuracy of the model only in the context of the same data set which has been employed during the training of the model. These in-domain assessments are not sufficient to capture real clinical conditions where imaging data can be received in various institutions, equipment vendors, protocols of acquisition and the demographics of the patient. These variations bring about distributional variations among datasets, which are often known as domain shift.

In case of domain shift, the training data and deployment data are characterized by different statistical properties, which can substantially reduce the model training. This problem is of primary concern in the field of medical imaging since medical decisions regarding patients can be influenced by predictions that are not reliable. Although the use of convolutional neural networks (CNNs) to detect pneumonia is increasing, little systematic comparison of the model generalizability to different mixed-based datasets of chest X-rays has been conducted.

More precisely, popular pretrained architectures, including VGG16, ResNet50 and DenseNet121, have been widely tested in single-dataset configurations. Although, it remains unclear:

- To what extent these architectures maintain performance when evaluated on unseen datasets.
- Whether certain architectures are inherently more robust to domain shift.
- How model interpretability, as measured through Grad-CAM visualizations, changes under cross-dataset evaluation.

Thus, the question that the work will focus on is as follows: To determine the extent to which domain shift affects pneumonia detection metrics and to determine the cross-dataset robustness of popular CNN architectures using quantitative performance and interpretability analysis.

Mathematically, assume that D_A and D_B are two chest X-ray data sets that are based on different distributions. The model that was trained on D_A is tested on D_A (in-domain) and D_B (cross-domain). The deterioration of performance between these two assessments serves as an indicator of domain shift effect. This is performed on both sets of data in a symmetric manner in order to evaluate generalization in either way.

The research paper aims to identify whether the high in-domain performance of Kaggle can be converted to robust cross-domain performance and what the architecture that is more robust can be by performing a systematic study of four experimental setups —Kaggle→Kaggle, Kaggle→RSNA, RSNA→RSNA, and RSNA→Kaggle.

Such an issue must be solved to achieve reliable and clinically implementable medical AI systems, in which cross-establishment across institutions is one of the major prerequisites.

III. RELATED WORK

Deep learning has greatly progressed automated pneumonia detection based on chest X-ray (CXR) images. The use of the Convolutional Neural Networks (CNNs) has proven to be very effective in high classification performance. Most of the existing literature however concentrates more on the single dataset evaluation with little emphasis made on the cross dataset evaluation, and domain shift analysis.

A. Pneumonia Detection Using CNNs

Wang et al. presented one of the first large-scale benchmarks on the topic of chest X-ray classification with the NIH ChestX-ray14 dataset. Their experiment showed that deep CNNs can be used to classify multi-labels of thoracic diseases, including pneumonia. Then, Kermany et al. applied transfer learning methods to datasets of pediatric chest X-rays (usually provided by Kaggle) and reported high accuracy in detecting pneumonia. These studies formed the baseline in using already trained architectures like VGG, ResNet, and DenseNet in medical imaging problems.

Rajpurkar et al. suggested the CheXNet, a 121-layer DenseNet model, which was trained on the NIH dataset, and claimed radiologist-level results in pneumonia detection. This paper exhibited the promise of deeper architectures with high levels of connectivity on medical image classification. Nonetheless, the evaluation was done on the same data, and cross-dataset robustness was not investigated.

B. Transfer Learning in Medical Imaging

The limited labeled medical datasets have also led to the predominance of transfer Learning. Popular pretrained models like VGG16, ResNet50 and DenseNet121, which were initially trained using the ImageNet data set, have been demonstrated to exhibit good performance when they are fine-tuned on chest X-ray images. There are several studies with better than 90 percent classification accuracies on the Kaggle Chest X-ray dataset. Although these results are promising, the majority of the evaluation can only be performed on the same dataset

through training and testing. Although high in-domain accuracy demonstrates learning ability, this does not as such predicts a generalization to images of images, which have been gathered in varying clinical conditions.

C. Domain Shift and Cross-Dataset Generalization

Domain shift is one of the key problems in medical AI. Zech et al. established that CNN models trained on pneumonia detection can be conditioned to hospital-specific features instead of detecting patterns related to pathology. Their results showed that there were major performance declines when models were tested across hospitals, underscoring the existence of spurious relationships. Likewise, Cohen et al. examined cross-dataset generalization in chest X-ray models and found that there was significant degradation when models were cross-trained to datasets of different protocols of acquisition. These works imply that deep learning models can overfit to artifacts predicting datasets, which makes the problem of their implementation in real-life questionable.

Previous studies, however, determine the occurrence of domain shift, and no systematic architectural comparisons were conducted in controlled cross-dataset environments. There has been limited work on a direct comparison of multiple CNN architectures trained on one data and evaluated on another in a bidirectional manner.

D. Model Interpretability in Pneumonia Detection

Gradient-weighted Class Activation Mapping (Grad-CAM) was one of the most common interpretability methods to visualize the model attention locations in chest X-ray classification. Selvaraj et al. developed Grad-CAM, which emphasizes the discriminative parts of images as with regards to CNN decisions. Grad-CAM has also been utilized in medical imaging to ensure that the models can be relied upon to identify clinically relevant regions in the lungs. Despite the popular application of interpretability methods to validate predictions, there is minimal research examining the behaviour of attention patterns during domain shift. To determine reliability it is important to understand whether model attention changes to irrelevant regions in cross-dataset evaluation.

E. Research Gap

Based on the literature review, there are three apparent gaps:

- Most pneumonia detection papers claim high in-domain accuracy without considering the cross- dataset generalisation.
- Comparison of the relative strengths of various pretrained CNNs across domain shift is underrepresented.
- Interpretability analysis is not frequently used in conjunction with cross-dataset analysis in order to study attention stability.

These gaps have been bridged by this work which conducts a cross dataset analysis of VGG16, ResNet50 and DenseNet121 in both Kaggle and RSNA chest X-ray datasets. Through both quantitative performance degradation and Grad-CAM attention behavior analysis, the research gives deeper

ideas about architectural robustness and deployability in the real world.

IV. METHODOLOGY

A. Overview of the Proposed Framework

The cross-dataset generalization of three pretrained Convolutional Neural Network (CNN) architectures in detection of pneumonia in chest X-ray (CXR) images is assessed in the proposed study. The entire pipeline is composed of:

- Dataset preprocessing and normalization
- Transfer learning using pretrained CNN models
- In-domain and cross-domain evaluation
- Performance metric computation
- Grad-CAM-based interpretability analysis

Two datasets with different distributions are used:

- Kaggle Chest X-Ray Dataset
- RSNA Pneumonia Detection Dataset

There are four experiment conditions analyzed:

- Kaggle → Kaggle (in-domain)
- Kaggle → RSNA (cross-domain)
- RSNA → RSNA (in-domain)
- RSNA → Kaggle (cross-domain)

This bi-directional configuration allows methodological measurement of domain shift effect.

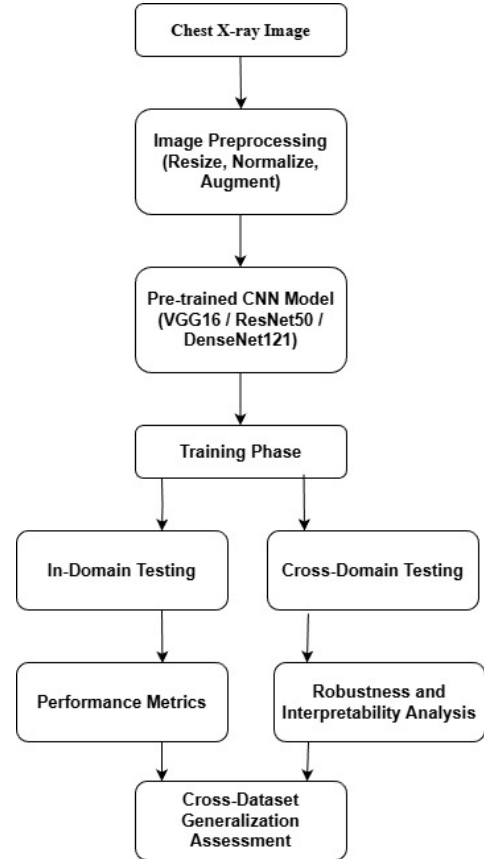


Fig. 1: Fig. 1: Overall Cross-Dataset Evaluation Framework

B. Dataset Preparation and Preprocessing

- Images resized to 224×224 pixels.
- Converted to RGB format.
- Pixel normalization to $[0,1]$ range.
- Dataset-specific train/test split.
- Data augmentation (training only): horizontal flipping, small rotation ($\pm 15^\circ$), zoom.
- No augmentation during testing.

C. CNN Architectures

The choice of three popular pretrained CNN architectures was made according to popular use in medical image classification:

1) *VGG16*: VGG16 is a 16-layer deep CNN model described by a series of sequential layers of 3×3 convolutional layers with subsequent max-pooling. It has a simple and unified architecture that is applicable to transfer learning tasks.

Key characteristics:

- 13 convolutional layers
- 3 fully connected layers
- Approximately 138 million parameters
- ReLU activation

In this paper, the last fully connected layer was substituted with:

- Global Average Pooling
- Dense layer (ReLU)
- Dropout layer (0.5)
- Output sigmoid layer (binary classification)

2) *ResNet50*: ResNet50 introduces residual connections that allow gradients to propagate through deep networks effectively.

Key characteristics:

- 50 layers
- Residual skip connections
- Bottleneck blocks
- Approximately 25 million parameters

The residual learning will prevent the vanishing gradient issue and enhance the deep network learning stability.

3) *DenseNet121*: DenseNet121 links each layer with all the following layers of dense blocks, which encourages the use of features, and better gradient flow.

Key characteristics:

- 121 layers
- Dense connectivity pattern
- Fewer parameters than VGG16
- Strong feature propagation

The high degree of connectivity eliminates redundancy and enhances generalization and thus a good candidate of cross-domain robustness.

D. Transfer Learning Strategy

ImageNet pretrained weights were used to initialize all of their models. The training plan involved:

- 1) Freezing early convolutional layers

- 2) Fine-tuning deeper layers

- 3) Substituting the head of classification by a custom binary output layer.

Training configuration:

- Optimizer: Adam
 - Learning rate: $1e-4$
 - Loss function: Binary Cross-Entropy
 - Batch size: 32
 - Epochs: 10–15 (early stopping applied)
- Overfitting was avoided through early stoppage.

E. Cross-Dataset Evaluation Protocol

In order to measure the domain shift, the following evaluation protocol was used. Assume D_A and D_B represent the Kaggle and RSNA datasets, respectively.

The models were trained on both D_A and tested on:

- D_A (in-domain performance)
- D_B (cross-domain performance)

Performance degradation was computed as:

$$\Delta = \text{Accuracy}_{\text{in-domain}} - \text{Accuracy}_{\text{cross-domain}} \quad (1)$$

The higher Δ means that it is more sensitive to domain shift.

F. Evaluation Metrics

The performance metrics based on the following were applied:

- Accuracy
- Precision
- Recall (Sensitivity)
- F1-Score
- Area Under ROC Curve (AUC)

Each experiment was also used to create confusion matrices.

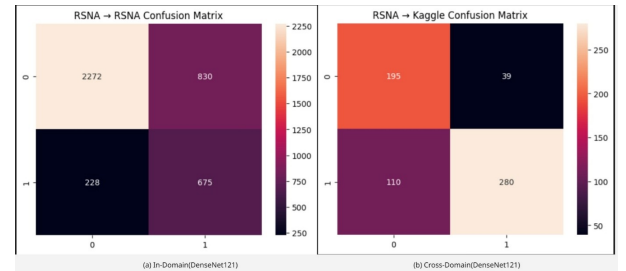


Fig. 2: Confusion matrix comparison for DenseNet121 under (a) in-domain and (b) cross-domain evaluation. Increased false predictions under cross-domain settings indicate domain shift impact.

G. Grad-CAM Interpretability Analysis

Grad-CAM (Gradient-weighted Class Activation Mapping) was used to test the images to examine model attention behavior. Grad-CAM produces heatmaps that indicate the discriminating areas of the image that are affecting classification decisions.

The comparisons were made as follows:

- In-domain attention maps
- Cross-domain attention maps
- Architecture-wise attention differences

This assessment can be used to find out whether models are devoted to clinically essential lung areas or data-specific artifacts with domain shift scenarios.

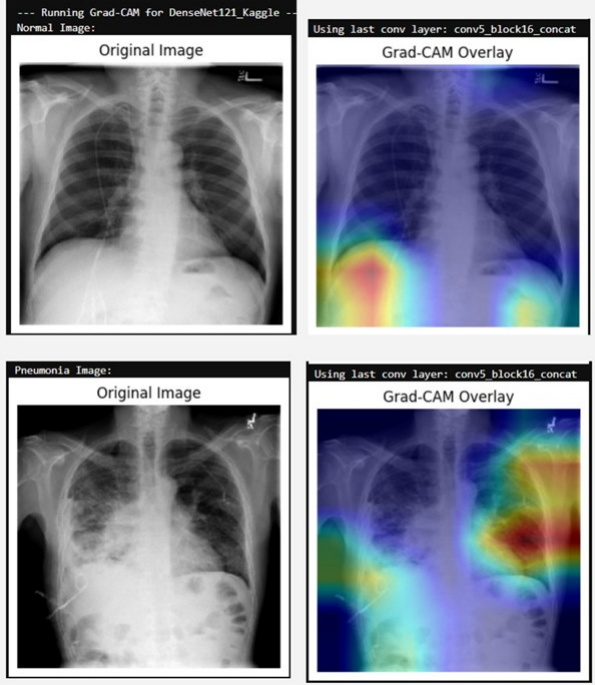


Fig. 3: Grad-CAM Visualization Comparison (Kaggle→Kaggle vs Kaggle→RSNA)

V. EXPERIMENTAL RESULTS

A. Overview

The quantitative analysis of VGG16, ResNet50 and DenseNet121 conducted in-domain and cross-domain experimental conditions is presented in this section. The main goal is the measurement of the degradation of the performance caused by domain shift and a comparison of architectural robustness. Four experimental settings have been tested:

- Kaggle → Kaggle
- Kaggle → RSNA
- RSNA → RSNA
- RSNA → Kaggle

Accuracy, Precision, Recall, F1-score and AUC were used to measure performance.

B. In-Domain Performance

Table I will provide the overall performance of all three architectures when trained and tested with the same dataset to classify images of pneumonia. This assessment is used as a baseline to determine the capability of each of the models

given ideal conditions with the training and testing data distributions aligned. The findings indicate relative superiorities of the architectures in revealing patterns of pneumonia without the effect of domain shift. This in-domain analysis provides a point of reference when other cross-dataset assessments are performed.

TABLE I: In-Domain Performance Comparison

Model	Dataset	Accuracy	Precision	Recall	F1-Score	AUC
VGG16	Kaggle	81%	91%	77%	84%	89%
ResNet50	Kaggle	57%	96%	33%	49%	83%
DenseNet121	Kaggle	87%	92%	87%	89%	92%
VGG16	RSNA	71%	42%	67%	51%	76%
ResNet50	RSNA	50%	29%	82%	42%	67%
DenseNet121	RSNA	73%	45%	75%	56%	81%

In both datasets, the high-performance in in-domain was most represented by DenseNet121 followed by VGG16 and ResNet50. This suggests that the pneumonia classification of deeper architectures with high connectivity is useful during training and testing distribution matching. Nevertheless, strong in-domain performance does not always imply strong generalization.

C. Cross-Dataset Performance

Table II shows the cross-dataset evaluation results.

TABLE II: Cross-Dataset Performance Comparison

Training	Testing	Model	Accuracy	Precision	Recall	F1	AUC
Kaggle	RSNA	VGG16	44%	28%	94%	44%	72%
Kaggle	RSNA	ResNet50	54%	29%	70%	41%	63%
Kaggle	RSNA	DenseNet121	45%	28%	90%	43%	70%
RSNA	Kaggle	VGG16	50%	98%	21%	34%	86%
RSNA	Kaggle	ResNet50	68%	91%	54%	68%	84%
RSNA	Kaggle	DenseNet121	76%	88%	72%	79%	85%

General performance drop was detected in all models during cross-dataset evaluation-testing, proving that domain shift occurred in between Kaggle and RSNA datasets. Out of the three architectures:

- DenseNet121 showed the smallest performance degradation.
- ResNet50 demonstrated moderate robustness.
- VGG16 exhibited the highest sensitivity to domain shift.

D. Quantifying Domain Shift Impact

To measure generalization loss, performance degradation (Δ) was computed as:

$$\Delta = \text{Accuracy}_{\text{in-domain}} - \text{Accuracy}_{\text{cross-domain}} \quad (2)$$

Table III summarizes the accuracy degradation.

Interpretation:

- Positive Δ → accuracy dropped in cross-domain → domain shift caused performance degradation.
- Negative Δ → cross-domain accuracy slightly higher than in-domain → model generalized well (ResNet50 and DenseNet121 in RSNA → Kaggle).

TABLE III: Accuracy Degradation Due to Domain Shift

Model	Kaggle $\rightarrow$ RSNA Δ	RSNA $\rightarrow$ Kaggle Δ
VGG16	37%	21%
ResNet50	3%	-18%
DenseNet121	42%	-3%

- DenseNet121 shows large degradation for Kaggle $\rightarrow$ RSNA, but small or negative for RSNA $\rightarrow$ Kaggle, confirming its robustness across datasets.

E. Confusion Matrix Analysis

The confusion matrices were used to learn about the class wise prediction behavior in case of domain shift.

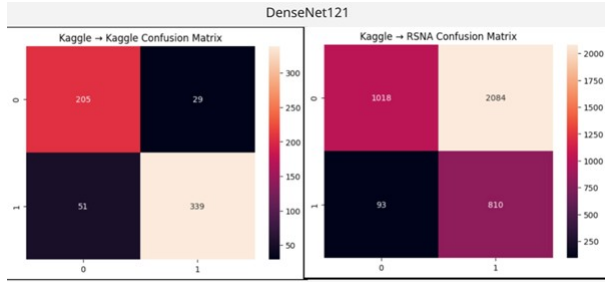


Fig. 4: Fig 4: Confusion Matrix Comparison (In-Domain vs Cross-Domain)

There is an increment in false positives and false negative results in cross-domain contexts. Cases of pneumonia were more often misclassified in cross-dataset assessments, and so identified learned pathological patterns might be dataset-specific.

F. ROC Curve Analysis

Receiver Operating Characteristic (ROC) curves also demonstrate the ability of models to be discriminated.

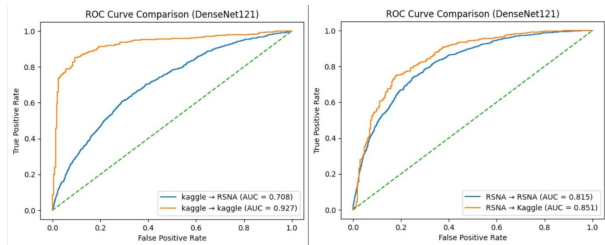


Fig. 5: Fig 5: ROC Curves for In-Domain and Cross-Domain Evaluations

The performance of denseNet121 on Kaggle (AUC = 0.927) was strong in-domain as at the same time it scored moderate on RSNA (AUC = 0.815). There was a severe degradation when the Kaggle-trained model was tested on RSNA (AUC = 0.708), which reflects strong domain shift. On the other hand, the RSNA-trained model exhibited a superior generalization to Kaggle (AUC = 0.851) such that RSNA training data could be more distributional diverse.

G. Grad-CAM Analysis Results

Grad-CAM visualisations show qualitative differences between in-domain and cross-domain attention maps.

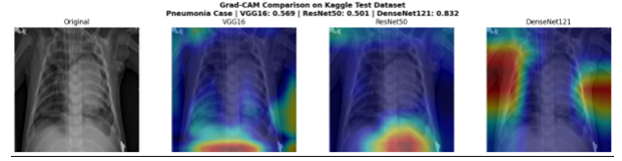


Fig. 6: Fig. 6: Grad-CAM Comparison Across Architectures

Key observations:

- In-domain predictions showed focused attention on lung opacity regions.
- Cross-domain predictions occasionally highlighted non-lung regions.
- DenseNet121 maintained more stable attention localization across domains.
- VGG16 exhibited inconsistent attention patterns under domain transfer.

H. Summary of Key Findings

The outcomes of the experiments prove that:

- High in-domain accuracy does not guarantee cross-domain robustness.
- Domain shift significantly impacts pneumonia detection performance.
- DenseNet121 is the most robust architecture among the evaluated models.
- Interpretability analysis confirms attention instability under domain transfer.
- Cross-dataset validation is essential for clinically deployable AI systems.

VI. DISCUSSION

A. Impact of Domain Shift on Pneumonia Detection

The experiment findings evidently prove that the deep learning models that have been trained on pneumonia detection are extremely sensitive to domain shift. Although the three architectures have shown great performance in terms of in-domain, a high level of degradation was realized in cross-dataset assessments. This observation supports the fact that high accuracy in single-dataset studies can be the overestimation of performance in the real world.

Kaggle and RSNA datasets vary in terms of the methods of image acquisition, distribution of resolutions, annotation, and patient demographics. These variations make changes in pixel-level statistics and feature distributions resulting in less generalization by models that are tested out of their training range. The general decrease in the performance in all the architectures confirms that domain shift is one of the main challenges when it comes to medical imaging AI.

Clinically, this is significant in terms of validating AI systems in multi-institutional datasets prior to implementation.

B. Architectural Robustness Analysis

DenseNet121 was the architecture that showed the least performance degradation when tested at cross-dataset transfer. This is indicating that high density of connection facilitates enhanced reuse of features and higher gradient transfer, which allows the model to acquire more transferable representations.

ResNet50 proved to be rather robust, presumably because of the training of the network under a residual mechanism, which stabilizes training of deeper networks. Nevertheless, VGG16 was the most sensitive to domain shift. VGG16 has a sequential architecture and may therefore not retain generalized feature representations as well as may be the case with skip or dense connection architectures.

These results show that architectural design is of critical importance in cross-domain generalization. The models with improved connectivity mechanisms are more appropriate to implement in the heterogeneous medical settings.

C. Interpretability Under Domain Shift

The analysis of Grad-CAM showed significant qualitative information. In in-domain settings, models were predominantly concentrated on areas of lung opacity that were related to pneumonia. Nevertheless, during the cross-dataset appraisal, occasionally, the attention map moved towards irrelevant parts of the image such as image edges or un-infected parts.

This model behavior also suggests a possible partial dependency of models on dataset-based artifacts and not necessarily on pure pathological characteristics. The instability of attention localization with change of domain leads to the question of the reliability of the model and clinical trustworthiness.

DenseNet121 also showed a relatively stable attention tendency in data sets, which also supports its strength. Conversely, the attention dispersion of VGG16 was higher when it was assessed with a new domain, which is consistent with its larger performance degradation. These results indicate that interpretability methods like Grad-CAM need to be employed as a tool of both qualitative validation and diagnostic to assess their robustness to changes in distribution.

D. Clinical Implications

The implications of the results of this study are substantial to the real world AI implementation:

- 1) Cross-dataset validation must be regarded as an essential assessment procedure in the research of medical AI.
- 2) The high single-dataset performance is not enough to assure some generalizability.
- 3) Quantitative measures should be achieved with interpretability analysis to achieve credible predictions.

Advanced generalization is an essential tool in avoiding diagnostic errors when the imaging devices and populations in clinical settings are interchangeable.

E. Comparison with Existing Literature

The major consideration of the previous research was to maximize in-domain accuracy. Even though there are works

that recognized domain shift, not many have performed systematic bidirectional cross-dataset comparison along with interpretability analysis.

The research builds on existing literature since it:

- Offering a systematic cross dataset evaluation protocol.
- Explicitly measuring performance decline.
- Comparison of architectural strength under identical conditions.
- Incorporation of Grad-CAM analysis to analyze the stability of attention.

The use of both the quantitative and qualitative assessment allows making this work a more complete evaluation of the pneumonia detection systems.

F. Key Insights

Through the discussion, the following insights have been obtained:

- Domain shift significantly impacts pneumonia detection accuracy.
- Architectural connectivity influences generalization capability.
- DenseNet121 is comparatively more robust to cross-domain transfer.
- Interpretability analysis reveals attention instability under distribution mismatch.
- Cross-dataset benchmarking is essential for clinically deployable AI systems.

VII. CONCLUSION

In this research, the cross-dataset generalization of three popular pretrained CNN models including VGG16, ResNet50, and DenseNet121 to identify pneumonia on the basis of chest X-ray images was examined. In contrast to the traditional studies, which consider performance on a single dataset, this study had a systematic analysis on the in-domain and cross-domain performance based on the Kaggle and RSNA chest X-ray datasets.

The experimental findings indicated that in-domain performance was high in all the models, but when cross-dataset-evaluation was performed, performance dropped dramatically. This validates the existence of domain shift across dataset and clarifies the drawback of single-dataset validation. The most robust architecture out of the assessed ones was DenseNet121 with the most minimal accuracy loss and more consistent Grad-CAM attention localization across the domains. It was found that ResNet50 was moderately robust and VGG16 the most sensitive in response to changes in the distribution.

Grad-CAM further indicated that domain shift has the potential to affect attention behavior which sometimes leads to models paying attention to areas that are not pathologic. This observation highlights the role of interpretability in determining the reliability of models.

On the whole, the results underline the fact that in-domain performance is not guaranteed to be reflected in the real world. Cross-dataset validation is an important requirement of an AI system which should be offered as clinically deployable. The

research helps make its contribution to the area as it gives a systematic approach to testing the robustness of architecture in a domain shift and incorporates the analysis of interpretability into cross-domain benchmarking.

VIII. FUTURE SCOPE

Although this research offers some important knowledge on the issue of cross-dataset generalization, there are still some gaps that can be explored in the future.

Initially, adding domain adaptation methods that include adversarial training, matching of consistencies of features or self-supervised pretraining may enhance resistance to cross-domain settings. Further studies can focus on domain adaptation without supervision or semi-supervision to mitigate the decline in performance in heterogeneous datasets.

Second, the evaluation should be extended to more multi-institutional data in order to have a more holistic view of generalization behavior. The test across various hospitals and imaging systems would be fine to model the real-life deployment scenarios.

Third, ensemble learning methods that utilize multiple architectures can be made more robust to distributional changes. The transferable feature learning may be further enhanced by studying hybrid models or attention-based mechanisms.

Fourth, quantitative assessment of stability of attention on saliency consistency measures may reinforce the interpretability assessment. Future work can provide quantifiable attention robustness scales as opposed to depending entirely on visual inspection.

Lastly, the integration of clinical metadata as age, gender, and patient medical history with the imaging characteristics could enhance better generalization performance and aid more consistent decision maker systems.

By considering such directions, the future studies can take another step toward creating reliable, strong, and clinically implementable AI systems to detect pneumonia and use AI in a wider range of application to medical imaging.

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